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Course Overview

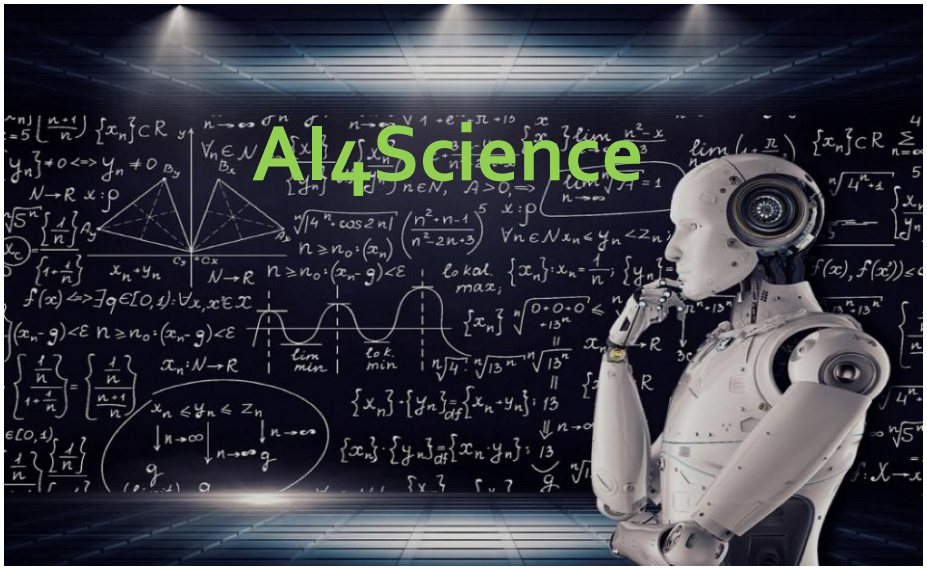
Statistical Machine Learning

Course Information 1/4

- **Instructor**
  - Dr. Junseok Kwon (권준석)
- **Lecture times and Venue**
  - Thursday 15:00-17:50
  - Room 209, 101

310 - 735

Course Information 1/4



Artificial intelligence (AI) and machine learning (ML) offer significant potential to revolutionize the fundamentals of scientific computation and discovery and AI is having a profound impact on science by accelerating discoveries across **physics, chemistry, biology, and engineering**.



LeCun argues that true machine intelligence must evolve beyond token prediction. Instead of solely focusing on language, future AI systems should embody: **Physical World Understanding** by representing and predicting continuous data streams from visual and sensory inputs.

Course Information 2/4

- **Scientific Machine Learning**
  1. How to integrate physical laws into data-driven learning
  2. How to design neural architectures that learn high-dimensional operators.

PDE → NN

2. **Physics-Informed Neural Networks (one PDE solution)**
   - MS-PINN, XPINN, Bayesian PINN, PINO

Continuous Continuous

3. **Neural Operator (Function → Function)**
   - DeepONet, GNO, LNO, MGNO, FNO

4. **Kolmogorov–Arnold Networks (Explicit Functional Structure)**
   - Kolmogorov–Arnold Theorem, Physics-Informed KANs

Course Information 3/4

- **Grade:** Project 90%
  - Apply Physics Law to your own research.
  - Write a paper that contains basic ideas (experimental results if possible).
- **Grade:** Participation 10%
  - Project Presentation (if you want).
- **Reference**
  - AI for Science Seminar (252-5256-00L)
  - AI in the Sciences and Engineering (2024)

Tentative Schedule 1/4

- **Scientific Machine Learning**
  - **Course overview 1주차 (03/04)**
  - **Neural PDE 1 2주차 (03/11)**
    - PDEs as learning targets
  - **Neural PDE2 3 주차 (03/18)**
    - Limitations of black-box neural networks
  - **PINNs 1 4 주차 (03/25)**
    - Physics-informed loss formulation, Forward and inverse PDE problems

Tentative Schedule 2/4

- **Scientific Machine Learning**
  - **PINNs 2 5주차 (04/01)**
    - Advanced PINNs
  - **Neural Operators 1 6주차 (04/08)**
    - Operator learning theory
  - **Neural Operators 2 7 주차 (04/15)**
    - DeepONet architecture
  - **Neural Operators 3 9 주차 (04/29)**
    - Advanced NOs

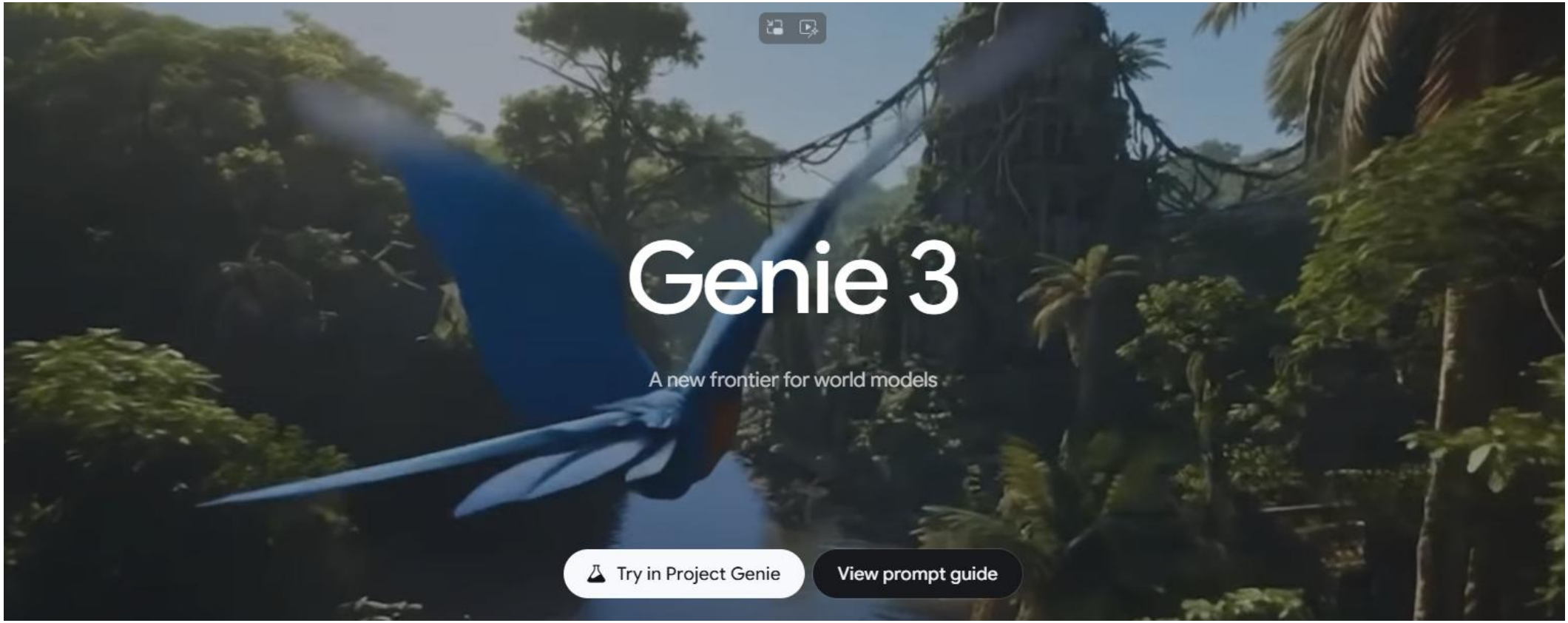
Tentative Schedule 3/4

- **Scientific Machine Learning**
  - **Kolmogorov–Arnold Networks 1 10주차 (05/06)**
    - Kolmogorov–Arnold representation theorem
  - **Kolmogorov–Arnold Networks 2 11주차 (05/13)**
    - KAN architecture and spline-based edges
  - **Kolmogorov–Arnold Networks 3 12 주차 (05/20)**
    - Advanced KANs
  - **Applications 13 주차 (05/27)**

Tentative Schedule 4/4

- Scientific Machine Learning
  - Project presentation 1 14주차 (06/03)
  - Project presentation 2 15주차 (06/10)

Application



https://www.youtube.com/watch?v=hUONc4FRS1Q